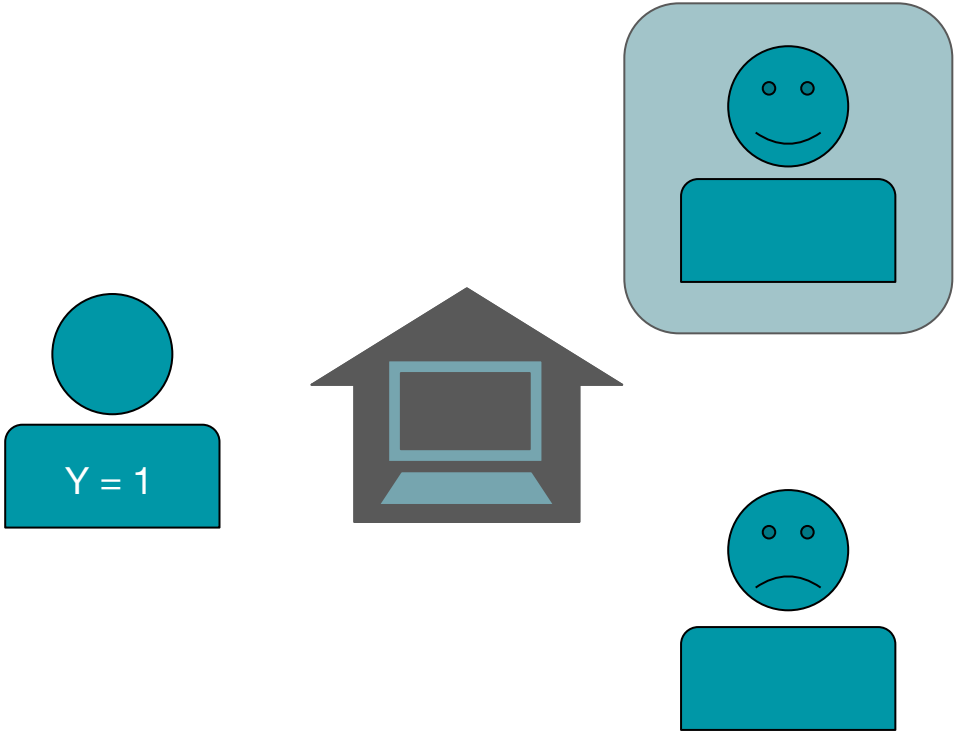
Fairness and Discrimination in Mechanism Design and Machine Learning

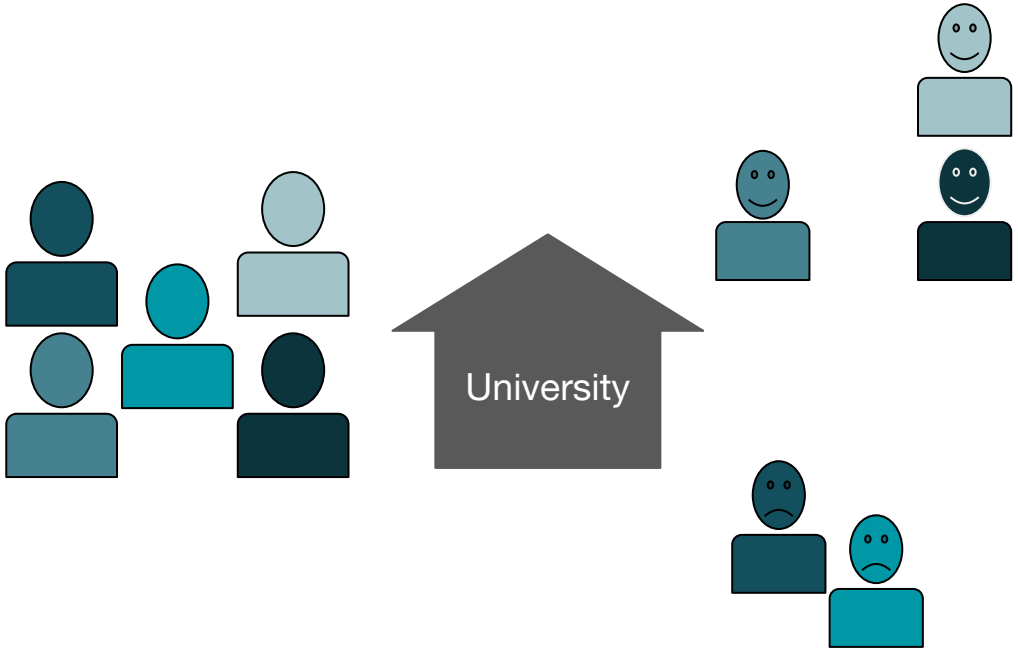
Jessie Finocchiaro[1], Roland Maio[2], Faidra Monachou[3]
Gourab K Patro[4], Manish Raghavan[5], Ana-Andreea Stoica[2], Stratis Tsirtis[6]

[1]Colorado
[2]Columbia
[3]Stanford
[4]IIT Kharagpur
[5]Cornell
[6]Max Planck Institute for Software Systems

Meet Jessie



But Jessie is just one person



Outline

- Setting
- Differences between Mechanism Design and Machine Learning
  - Perceptions of fairness and discrimination
- Lessons Learned
  - ML → MD
  - MD → ML

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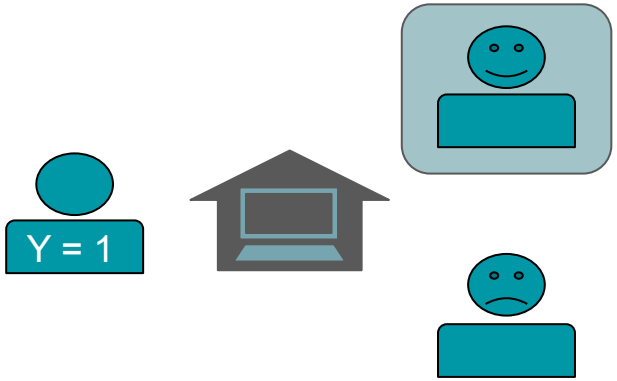
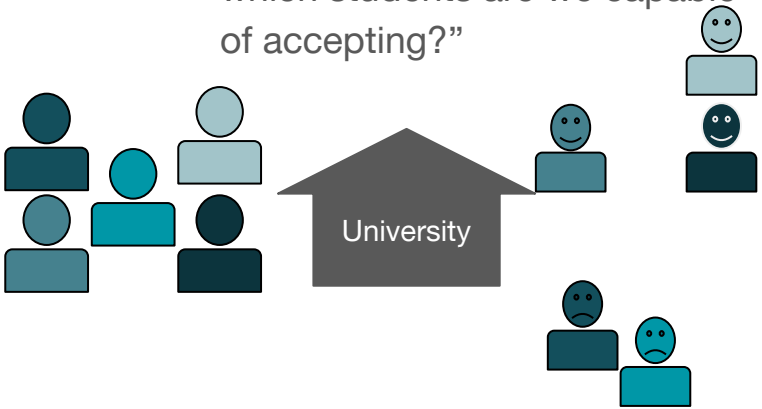
Setting

- Machine Learning
  - Often supervised, “true” observable label that we want to predict
  - Typically used in prescriptive settings
    - “Is Jessie qualified to attend University?”



- Mechanism Design
  - Agents are given payoff as a function of population decisions
  - Typically used in resource allocation settings
    - “Given our capacity constraints, which students are we capable of accepting?”



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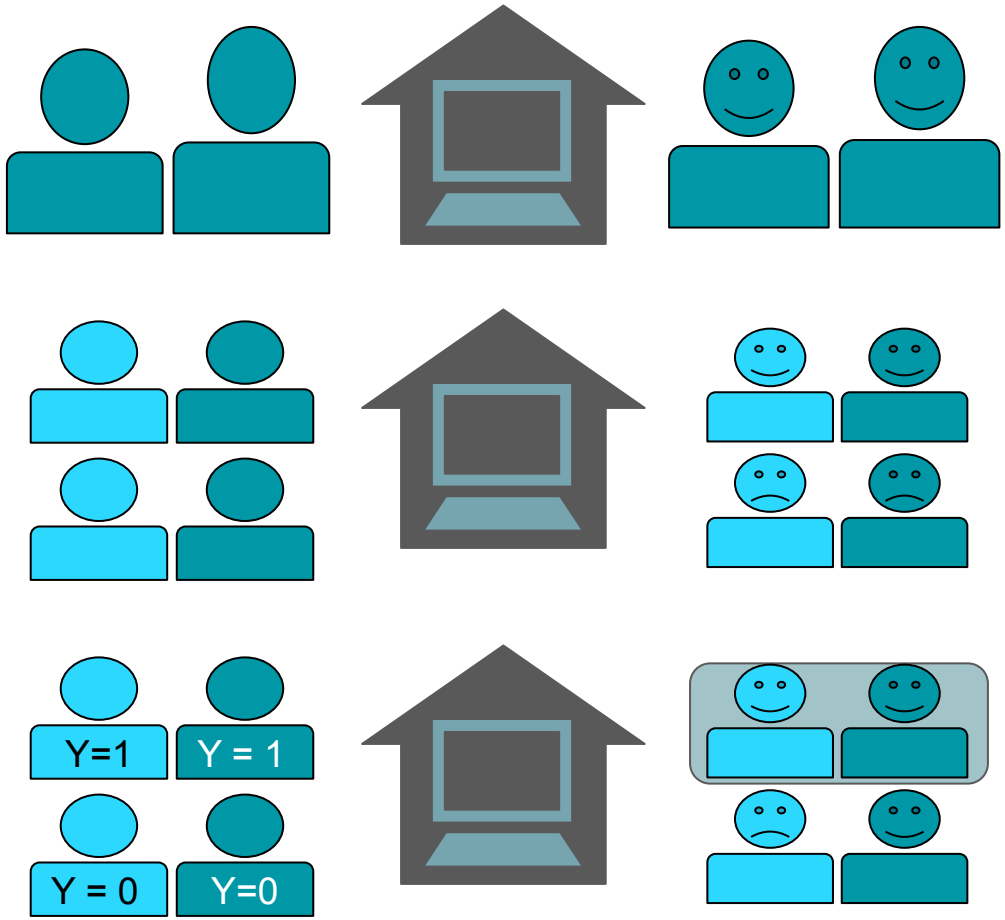
Fairness in ML

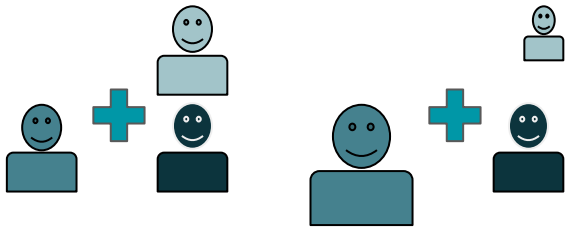
- What does “fair” even mean?
  - Individual Fairness
  - Group Fairness
    - Demographic Parity
    - Equalized Odds
  - The list goes on...



Fairness (and discrimination) in MD

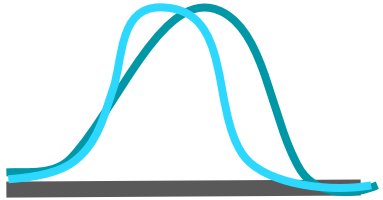
Fairness ≠ not discrimination

Fairness

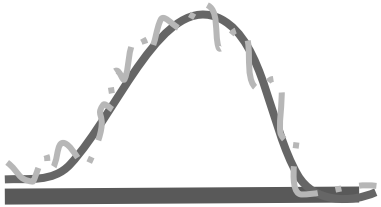


Discrimination

Taste-Based



Belief-Based



Statistical Discrimination
Coordination Failure
Mis-specification

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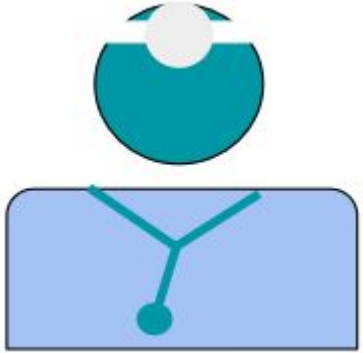
Lessons: ML → MD

(Re)defining notions of fairness

Mehrabi et al., 2019

Group-level diagnosis



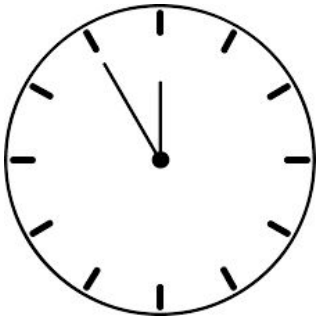
Abebe et al., 2020

Lessons: MD → ML

Tension between fairness and welfare

Kaplow and Shavell, 2003
Hu and Chen, 2018

Long-term effects of fairness



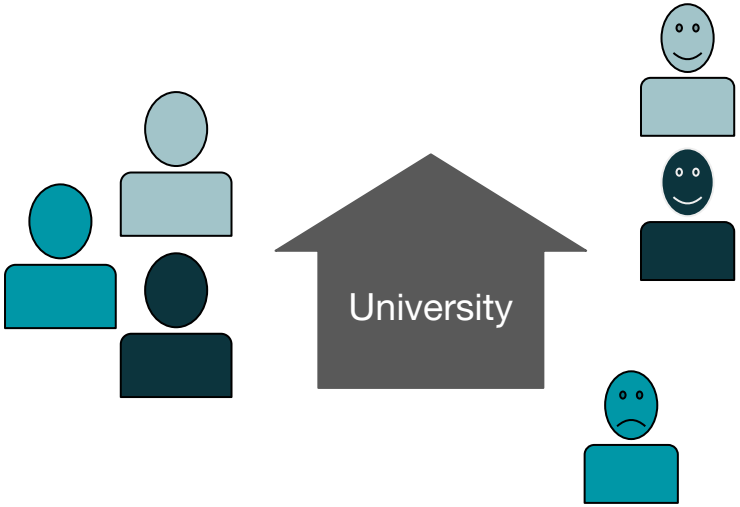
Liu et al. 2018
Kannan et al. 2019
Wen et al., 2019

Strategic agents

Hu et al., 2018

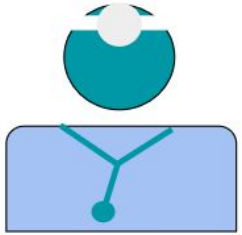
Summary: Why should anyone care about MD ∩ ML?

Criminal justice

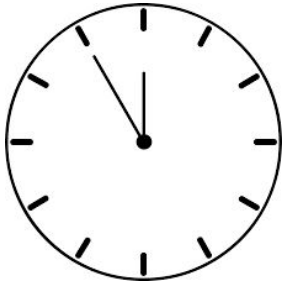


College admissions

Finance and lending



Healthcare



Population dynamics

Thank you!

University of Colorado Boulder

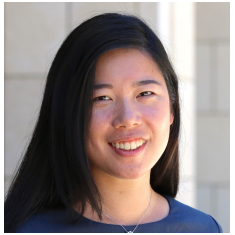
Columbia University

Columbia University

MAX PLANCK INSTITUTE FOR SOFTWARE SYSTEMS

Acknowledgements:

MD4SG
Mechanism Design for Social Good



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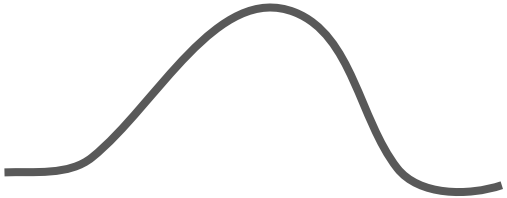
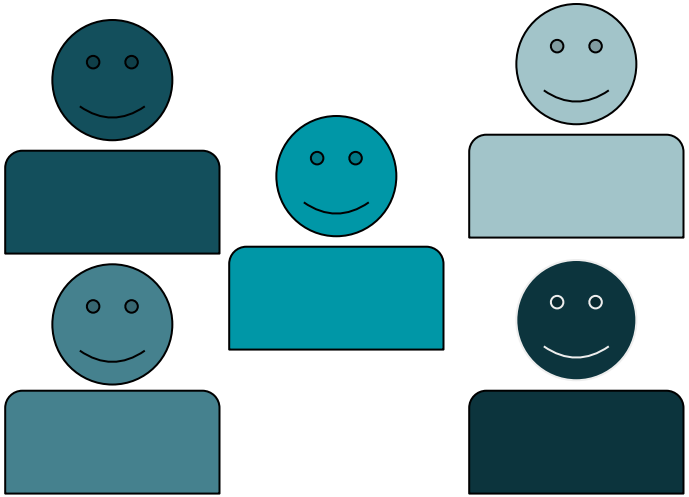
Application domains: unique challenges

- Criminal justice
  - High cost of false positive
- Advertising
- Education
- Labor markets and gig economy
- Healthcare
- Finance and lending
- Social networks





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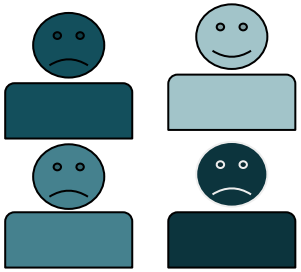

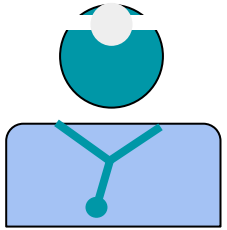

University